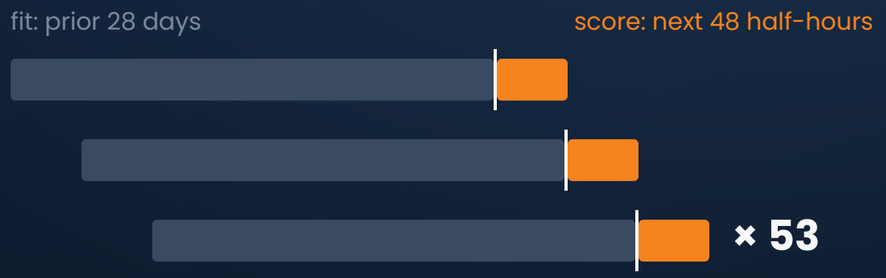
LOWER ERROR.
OR JUST NOISE?



white tick = origin, 00:00 London; refit from scratch at every origin

WALK-FORWARD

IN APP

$$\hat{\theta}_k = \mathrm{fit}\big(y_{[o_k-1344,\ o_k)}\big) \ \Rightarrow\ \hat{y}_{[o_k,\ o_k+48)}, \qquad o_{k+1} = o_k + 48$$

Refit each midnight on the prior 28 days; targets never enter the fit.

53 midnight origins: each lead is one clock slot

SCALED ERROR (MASE)

IN APP

$$\mathrm{MASE} = \frac{1}{53}\sum_{k=1}^{53} \frac{\mathrm{MAE}_k}{\overline{|y_t - y_{t-48}|}_{\text{ prior 28 days}}}$$

Each day's error over the prior 28 days' typical day-on-day change: unit-free.

ETS 0.84 · next best 0.89 · same slot yesterday 1.10

DIEBOLD–MARIANO

IN ENGINE

$$\mathrm{DM} = \frac{\bar{d}}{s_d/\sqrt{n}} \overset{H_0}{\approx} t_{n-1}, \qquad d_k = \mathrm{MASE}_k^{\mathrm{ETS}} - \mathrm{MASE}_k^{\mathrm{next}}$$

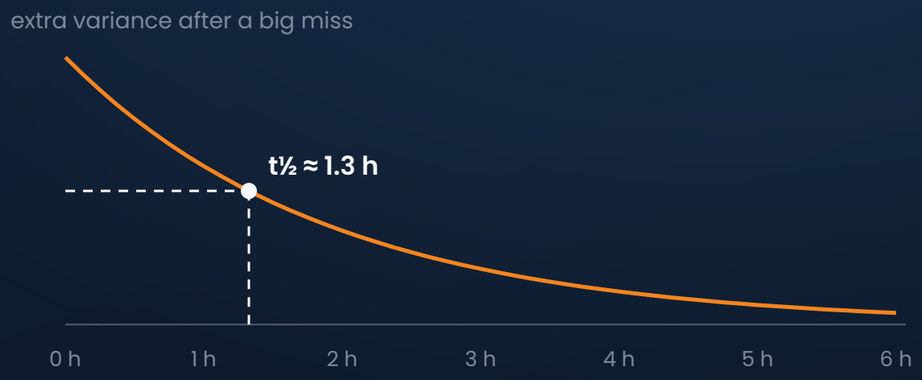
Is the daily gap bigger than day-to-day noise? Days don't overlap, so h = 1.

p = 0.02 over 53 days: unlikely to be noise

IN APP runs live in Eisengrau IN ENGINE coded and tested, not yet live

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ERRORS CLUSTER. CONSTANT-VARIANCE BANDS MISLEAD.



INTERVAL SCORE

IN APP

$$\mathrm{IS}_{80} = \frac{1}{n}\sum_t \left[(u_t - l_t) + 10\,(l_t - y_t)^+ + 10\,(y_t - u_t)^+\right]$$

Charges band width plus misses; coverage alone rewards bands that are too wide.

IS_{80} £/MWh: naive 508 (99% inside) vs mean 260 (76%)

ARCH-LM TEST

IN APP

$$e_t^2 = a_0 + \sum_{j=1}^{12} a_j\, e_{t-j}^2 + \eta_t, \qquad \mathrm{LM} = (n-12)\,R^2 \overset{H_0}{\approx} \chi^2_{12}$$

Do big misses follow big misses within 6 h?

LM = 196 vs 21 at 5%; 219 with time of day removed

GARCH(1,1)

IN ENGINE

$$\sigma_t^2 = \omega + \alpha_1 e_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad t_{1/2} = \frac{\ln 2}{-\ln(\alpha_1+\beta_1)}$$

Variance jumps after a big miss, then decays; the half-life says how fast.

Half-life ≈ 1.3 h, time of day removed (derived)

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WHAT PRICE REMEMBERS.

Next: residual-demand forecasts as model inputs, versioned at each origin.



AUTOCORRELATION

IN APP

$$\mathbb{E}\,(y_t - y_{t-k})^2 = 2\sigma^2(1-\rho_k) \;<\; \sigma^2 \iff \rho_k > \frac{1}{2}$$

Derived: the last price beats the 28-day mean only while ρ > ½.

0.81 at 30 min, 0.31 at 24 h: slide 5's winners

KALMAN FILTER

IN ENGINE

$$y_t = \mu_t + \gamma_t + \varepsilon_t, \quad \mu_{t+1} = \mu_t + \eta_t, \quad z_t = \frac{y_t - \hat{y}_{t|t-1}}{\sqrt{F_t}}$$

Splits price into a drifting level, a daily cycle and noise; z measures surprise.

|z| > 3 in 1.3% of half-hours; a normal gives 0.27%

DISTURBANCE SMOOTHER

IN ENGINE

$$r_t^{\varepsilon} = \frac{\hat{\varepsilon}_t}{\sqrt{H - \mathrm{Var}(\varepsilon_t \mid Y_n)}}, \qquad r_t^{\eta} = \frac{\hat{\eta}_t}{\sqrt{Q - \mathrm{Var}(\eta_t \mid Y_n)}}$$

In hindsight: one-off spike (ε) or lasting level shift (η)? Upstream denominator fixed.

20 spikes, 40 shifts flagged; ~7 each if the model held

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